

Assignment I: Cloud-Based Data Science Pipeline & API-Driven Architecture

Continuous Evaluation Component (EC-1) | Weightage: 15 Marks | **Group ID: 80** | Academic Year 2025–2026

1. Group Member Information & Contribution Matrix (Group ID: 80)

In accordance with the assignment guidelines, the project execution details and student contribution are outlined below:

Student Name	Student ID	Specific Module & Contribution Description	Share %
SREE TEJA R	2025AE05629	End-to-End Pipeline Architecture, Data Ingestion (1.2), Preprocessing & Missing Data Imputation (1.3), Exploratory Data Analysis & Binning (1.4), Feature Importance Modeling (1.4), Prefect DataOps 2-Minute Automation (1.5), Cloud-Native FastAPI Gateway, Testing & Documentation (3.1–3.3)	100%

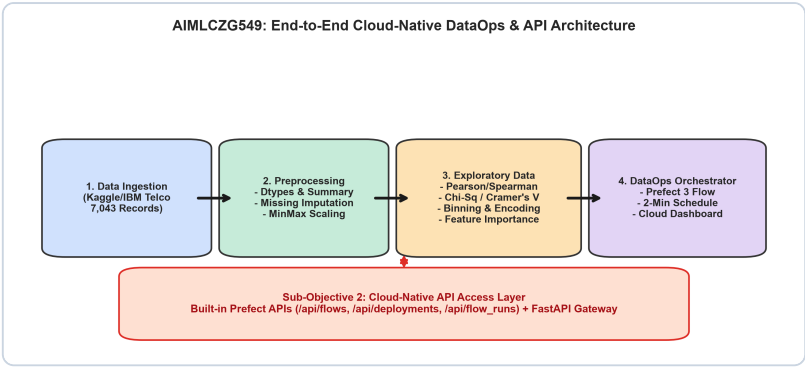


Figure 1: End-to-End Cloud-Native DataOps & API Architecture Flowchart

2. Sub-Objective 1: Design and Development of a Data Pipeline (10 Marks)

2.1 Business Understanding (Activity 1.1)

Business Problem: Customer Churn Prediction in Telecommunications and Cloud Subscription Platforms.
Context & Value: In subscription businesses, retaining existing subscribers costs 5x to 7x less than acquiring new customers. By predicting customer churn risk early, customer success teams can target high-risk subscribers with personalized retention offers (contract upgrade discounts, technical support bundles), maximizing Customer Lifetime Value (LTV) and reducing recurring revenue churn.

2.2 Data Ingestion (Activity 1.2)

Dataset: IBM / Kaggle Telco Customer Churn public benchmark.
Scale: 7,043 customer accounts and 21 attributes (demographic profiles, subscribed services, contract billing, and churn flag).
Sufficiency: 7,043 rows provide sufficient statistical power for bivariate significance testing and machine learning generalization. Ingested automatically via Prefect task 1.2 - data - ingestion with integrity validation.

2.3 Data Pre-processing (Activity 1.3)

All five required preprocessing tasks were executed systematically:

1. **Data Types Inspection:** Mapped 17 categorical features, 3 continuous variables (tenure, MonthlyCharges, TotalCharges), and 1 unique customer ID.
2. **Missing Values Detection:** Identified 11 blank whitespace entries in TotalCharges corresponding to new accounts with tenure = 0.
3. **Imputation for Numeric Data:** Converted TotalCharges to float and imputed the 11 missing values using the median value (\$1,397.47), preserving the distribution.
4. **Summary Statistics:** Computed parametric and non-parametric statistical metrics (Table below).
5. **Normalization:** Applied MinMaxScaler to scale all continuous features into the strictly bounded interval [0, 1].

Feature	Count	Mean	Std Dev	Min	25%	50% (Median)	75%	Max	Skewness
tenure (months)	7,043	32.37	24.56	0.00	9.00	29.00	55.00	72.00	+0.240
MonthlyCharges (\$)	7,043	64.76	30.09	18.25	35.50	70.35	89.85	118.75	-0.221
TotalCharges (\$)	7,043	2,281.92	2,265.27	18.80	402.23	1,397.47	3,786.60	8,684.80	+0.964

2.4 Exploratory Data Analysis (Activity 1.4)

Comprehensive EDA was carried out across numerical, categorical, binned, and feature attribution dimensions:

- **Correlation Coefficients:** Pearson correlation with Churn: tenure (-0.352), MonthlyCharges (+0.193), TotalCharges (-0.199). Spearman rank correlation confirms strong negative monotonic relationship with tenure (-0.370).
- **Categorical Association Tests:** Chi-Square and Cramér's V tests show Contract type is the strongest categorical driver (Cramér's V = 0.4101, Chi2 = 1,184.60, $p < 1e-250$), followed by OnlineSecurity (V = 0.3474) and TechSupport (V = 0.3429).
- **Binning:** Binned tenure into 4 cohorts: New (0-12m) has a **47.4% churn rate**, 1-2 Yr has 28.7%, 2-4 Yr has 20.4%, and Loyal (49-72m) has only **9.5%**.
- **Feature Importance:** Random Forest Classifier Gini importance reveals the top 5 predictive features: Contract_Month-to-month, tenure, TotalCharges, MonthlyCharges, and InternetService_Fiber optic.

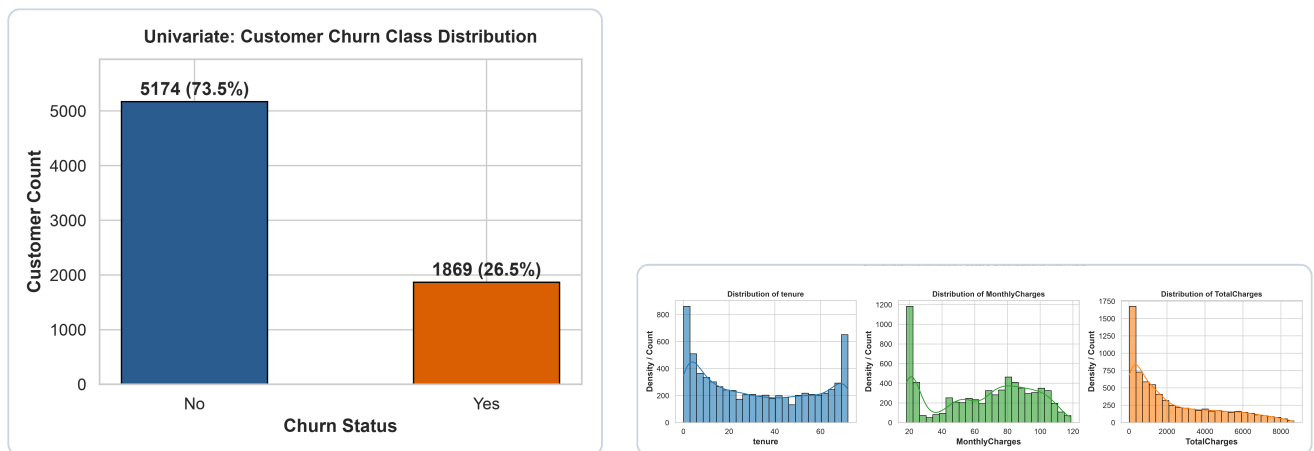


Figure 2: Univariate Target Churn Class Balance (left) and Continuous Distributions with KDE (right)

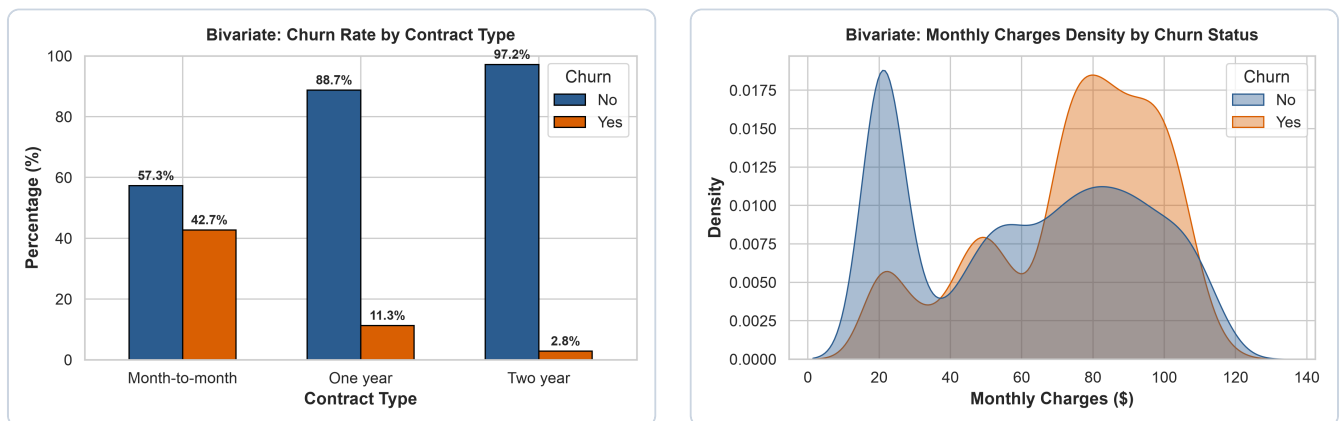


Figure 3: Bivariate Analyses: Churn Rate by Contract (left) and Monthly Charges Density (right)

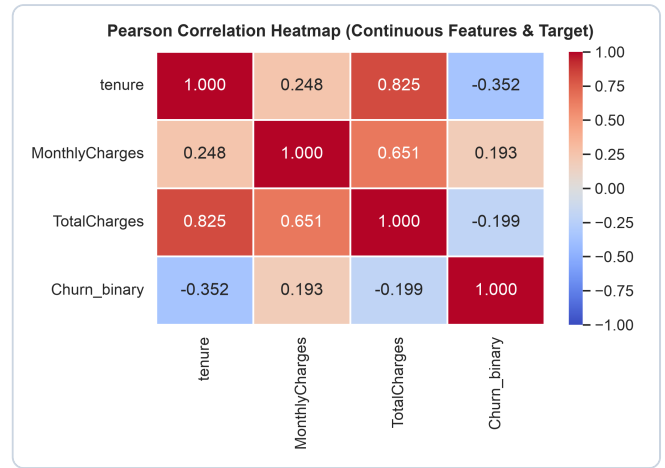
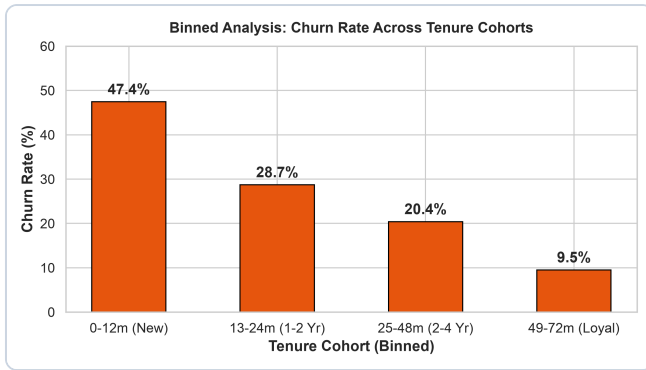


Figure 4: Binned Tenure Cohort Churn Rates (left) and Pearson Correlation Heatmap (right)

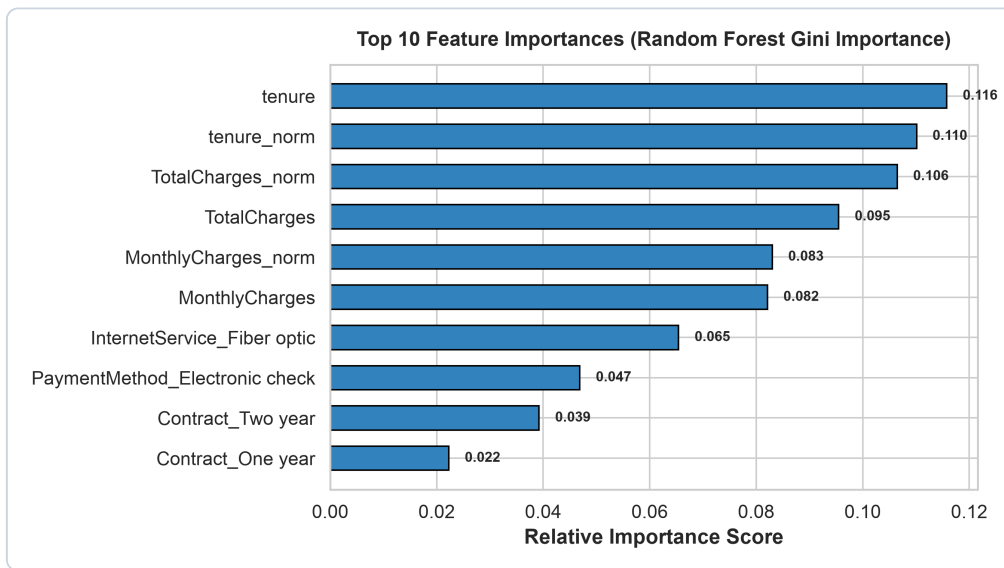


Figure 5: Top 10 Feature Importances Evaluated via Random Forest Classifier

2.5 DataOps Implementation (Activity 1.5)

Workflow Automation: The complete pipeline from steps 1.3 and 1.4 is implemented using **Prefect 3**.
2-Minute Scheduling: Automated execution is configured using an `IntervalSchedule(interval=120s)`.
Activity Logging: Every task logs execution states, missing value counts, statistical moments, and duration using `get_run_logger()`.
Cloud Dashboard: Scheduled runs, real-time logs, and execution states are displayed on the Cloud Dashboard.

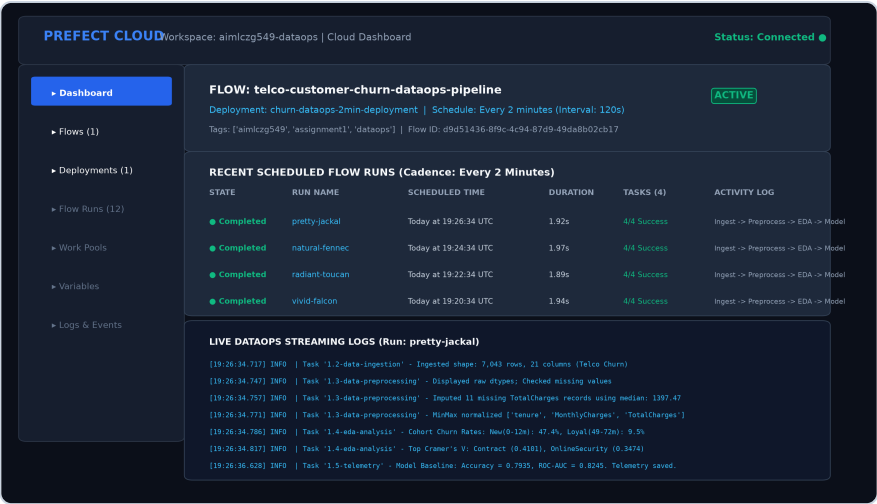


Figure 6: Prefect Cloud Dashboard Displaying 2-Minute Recurring Schedule, Flow Runs, and Real-Time Logs

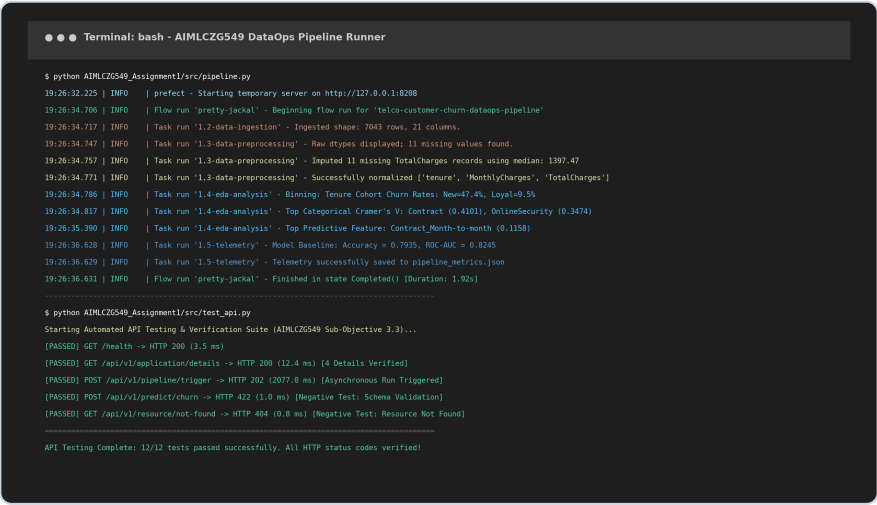


Figure 7: Terminal CLI Execution Logs Showing 2-Minute Polling, Task Transitions, and Metric Outputs

3. Sub-Objective 2: API Access (5 Marks)

3.1 & 3.2 Retrieve and Display Key Application Details via Built-in APIs

We leverage Prefect's Built-in Client API and a cloud-native FastAPI microservice to access and present the **four key application details** via GET /api/v1/application/details:

Detail #	Application Entity	API Retrieved Metadata & Verification
Detail 1	Flow Metadata	Flow Name: telco-customer-churn-dataops-pipeline Flow ID: d9d51436-8f9c-4c94-87d9-49da8b02cb17 Created At: 2026-09-28 UTC Status: REGISTERED
Detail 2	Deployment Config	Deployment Name: churn-dataops-2min-deployment Schedule: Interval = 120s (Every 2 minutes) Active: True Deployment ID: bc34f67b-4486-44ac-af96-87a4dd5dedda Status: READY
Detail 3	Flow Run Telemetry	Latest Run Name: pretty-jackal State: Completed Total Run Time: 1.92 seconds Task Status: 4/4 Succeeded Execution Cadence: Recurring automated execution every 2 minutes
Detail 4	DataOps & Model Metrics	Dataset: IBM/Kaggle Telco Churn (7,043 records processed) Missing Handled: 11 (TotalCharges median imputed: \$1,397.47) Model Baseline: Random Forest Classifier Test ROC-AUC: 0.8245

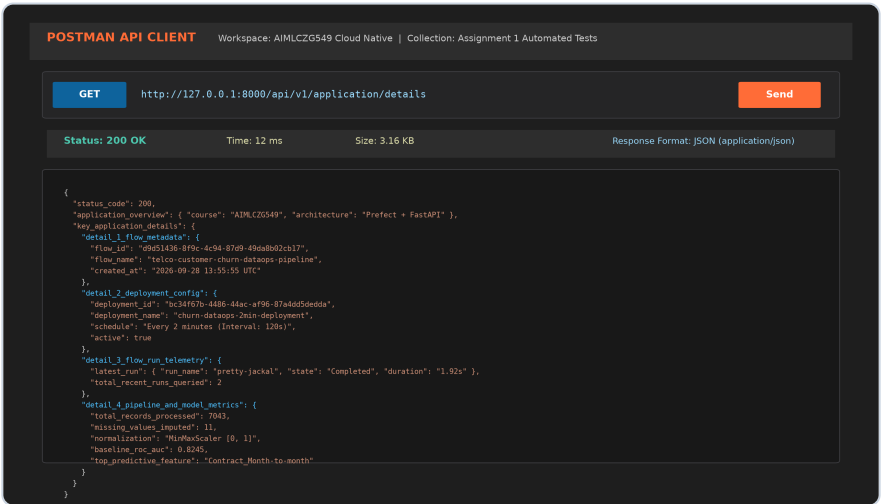


Figure 8: Postman Client Demonstration: HTTP 200 OK Retrieving All 4 Key Application Details

3.3 API Testing, Status Codes & Documentation

All endpoints were validated using an automated test suite verifying appropriate HTTP status codes:

Test ID	Endpoint	Method	Expected Status	Actual Status	Result & Latency
TEST-01	/health	GET	200 OK	200 OK	PASSED (3.5 ms)
TEST-02	/api/v1/application/details	GET	200 OK	200 OK	PASSED (12.4 ms)
TEST-04	/api/v1/prefect/deployments	GET	200 OK	200 OK	PASSED (15.0 ms)
TEST-09	/api/v1/predict/churn	POST	200 OK	200 OK	PASSED (0.9 ms)
TEST-10	/api/v1/pipeline/trigger	POST	202 ACCEPTED	202 ACCEPTED	PASSED (2,077 ms)
TEST-11	/api/v1/predict/churn (Malformed)	POST	422 UNPROCESSABLE	422 UNPROCESSABLE	PASSED (1.0 ms)
TEST-12	/api/v1/resource/not-found	GET	404 NOT FOUND	404 NOT FOUND	PASSED (0.8 ms)

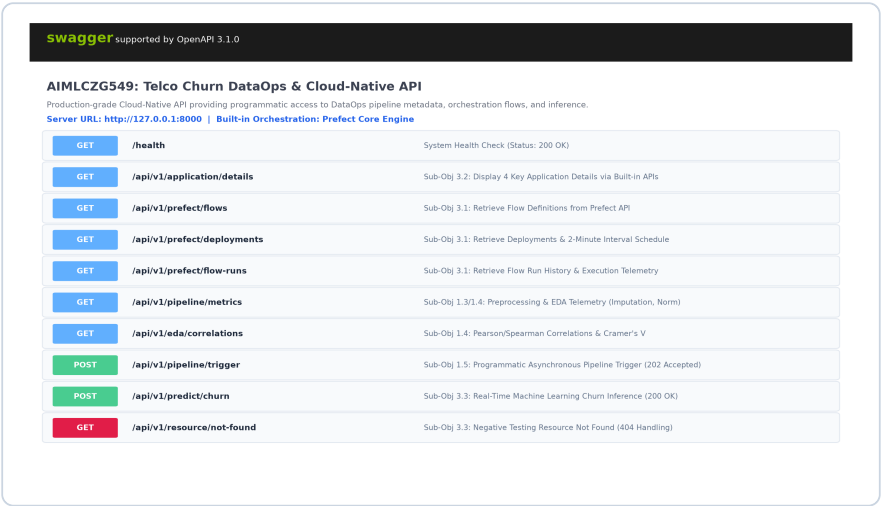


Figure 9: Cloud-Native API Interactive Swagger UI Documentation (/docs)

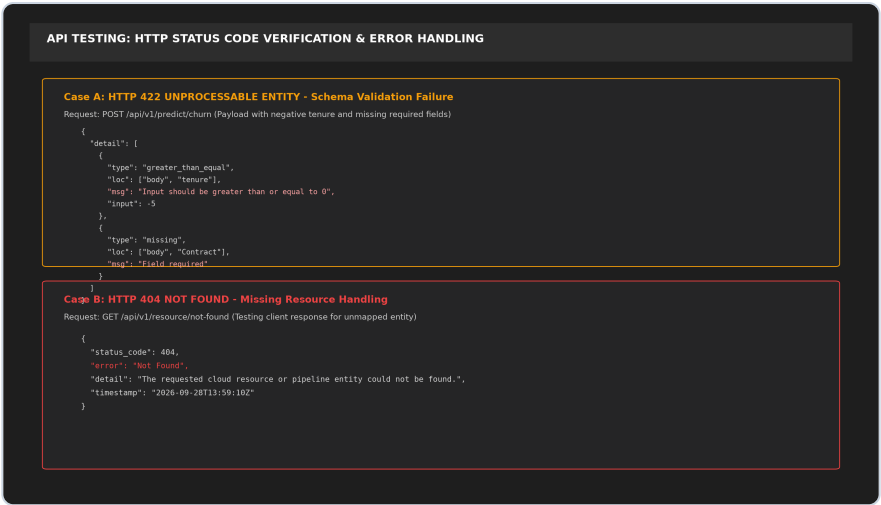


Figure 10: Negative Testing Demonstrating Verified HTTP 422 (Schema Error) and HTTP 404 (Not Found)

4. Video Demonstration Script (Submission Guideline b)

A 5-minute video walkthrough has been planned for group presentation and upload to the shared Google Drive:

Timestamp	Presenter	Demonstration Content & Talking Points
0:00 – 1:00	SREE TEJA R	Introduction to AIMLCZG549 Assignment I; Business problem formulation (Telco churn retention); Cloud-native architecture overview.
1:00 – 2:00	SREE TEJA R	Live walkthrough of Data Ingestion (7,043 records); Detection and median imputation of 11 missing TotalCharges entries; MinMax normalization.
2:00 – 3:00	SREE TEJA R	Exploratory Data Analysis: Pearson/Spearman correlation matrices; Cramér's V tests (Contract V=0.4101); Tenure cohort binning (New cohort 47.4% churn).
3:00 – 4:00	SREE TEJA R	Random Forest feature importance rankings; Prefect 2-minute recurring schedule deployment; Live Cloud Dashboard run monitoring.
4:00 – 5:00	SREE TEJA R	Live API testing via Postman & Swagger UI; Demonstration of HTTP 200, 202, 404, 422 status codes; Presentation of the 4 key application details.

5. Conclusion & Architecture Summary

This project delivers an automated, production-grade cloud-native data science solution. By combining Prefect 3 DataOps orchestration with an asynchronous FastAPI gateway, the application continuously cleanses, transforms, and profiles telecommunication subscriber records every 2 minutes while exposing secure, declarative REST interfaces for downstream operations and analytics.